Memory Management

subtitle

1s2026

Gabriela Bittencourt, João Pedro Leôncio and Vinícius Peixoto

Contents

1 Understanding Resources

Memory x Storage

When we talk about ‘memory’ in OS context, we are talking about **RAM**

Memory is NOT Storage

- Memory
  - fast, expensive, volatile
- Storage
  - slow, cheap, consistent

Program x Process

- Program
  - Code. Static entity.
  - Needs storage
  - Example: Your C code from MC202: the .c file
- Process
  - The execution of the code. Dynamic entity.
  - Needs memory: the instructions will be load in specific space of memory (), variables will be set to specific spaces in memory ()
  - Example: The instance of your text editor while you are writing a program; the instance of the terminal running when you are navigating through it to compile your code and when you run it; the instance of your program during the short period of time your computer is executing your code

Memory Management Historical Advance

- Contiguous Memory Allocation (no abstraction of memory)
  - One part is reserved to the OS, the other runs a single process
  - One process at the time: in order to change the process executed, the OS must save the previous process in disk and then load the next one.
  - Simple memory management
  - Poor memory utilization
- Non-contiguous Memory Allocation
  - a

Non-contiguous Memory Allocation

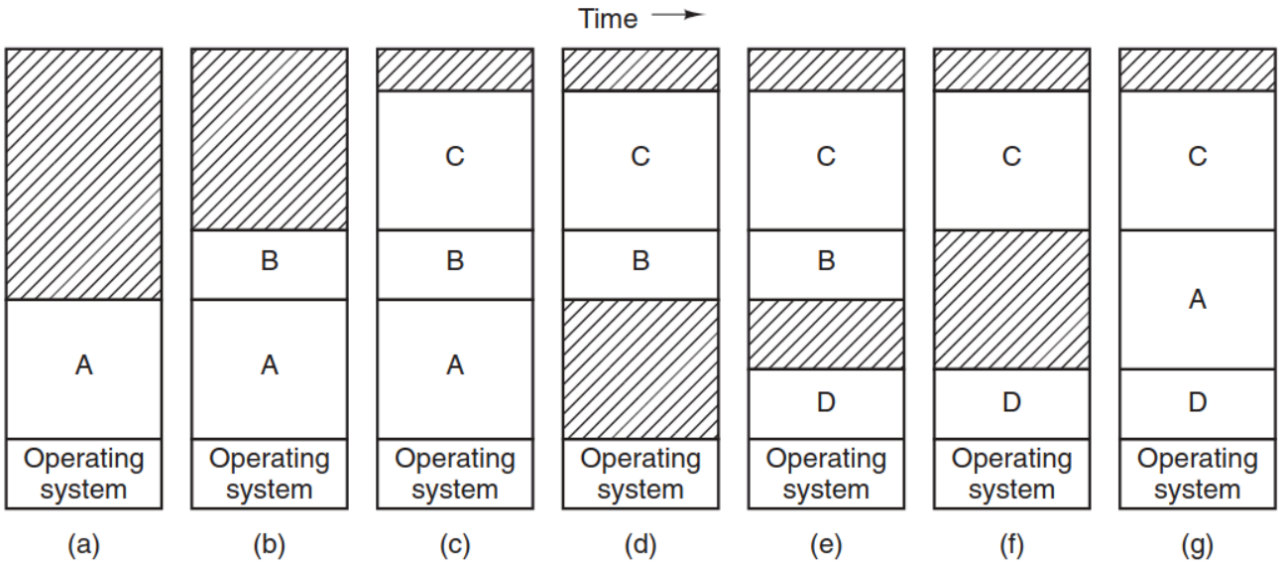

Figure 1: Memory allocation of 5 processes (programs: A, B, C, D and A again) over time.

2 Physical Memory

Limited resource

```
$ free -h
               total        used        free      shared  buff/cache   available
Mem:            14Gi        12Gi       803Mi       4.5Gi       6.2Gi       2.7Gi
Swap:          4.0Gi       2.6Gi       1.4Gi
```

3 Virtual Memory

Motivation

- Resource management
  - Ideally all processes would have access to the full memory at the moment of execution
  - In reality, each process has a access to a limited amount of memory that can expand and shrink depending on demand
- Scope security
  - security of data from different processes

4 Paging

Page Table Entries (PTE)

pte

Levels

pte is cool and all, but it's not enough

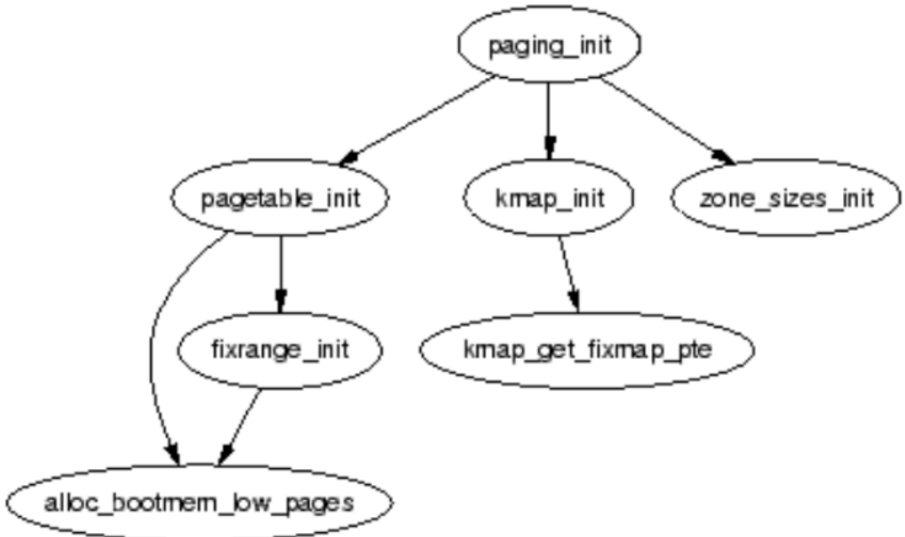

Figure 2: a - https://www.kernel.org/doc/gorman/html/understand/understand006.html

5 Struct

6 References

References

- Tanenbaum, Andrew. (2012) '3. Memory Management', in *Modern operating systems*. Pearson Education, Inc., 4th ed.

- TODO: add kernel codes
- TODO: add kernel docs